Generative Adversarial Networks and Summary of Course

FYS5429/9429

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May 21, 2026

Outline I

Outline II

Generative models: the common thread

All deep generative models share one goal: learn a distribution $p_{\boldsymbol{\theta}}(\boldsymbol{x})$ over data $\boldsymbol{x}$ that we can both evaluate and sample from.

Family	**Latent**	**Training signal**	**Sample cost**
VAE	$\boldsymbol{h} \in \mathbb{R}^{d_h},\ d_h \ll d_x$	ELBO	1 decoder pass
Diffusion	$\boldsymbol{x}_t \in \mathbb{R}^{d_x}$ (full dim)	**ELBO, T levels**	T denoiser calls
GAN	$\boldsymbol{z} \in \mathbb{R}^k$	Adversarial	1 generator pass

Key insight

Diffusion models are **hierarchical VAEs** with T latent levels, a fixed (non-learned) encoder, and full-dimensional latents. Understanding VAEs is the fastest route to understanding diffusion.

Transition: From Likelihood to Adversarial Training

So far we studied models maximising (a lower bound on) $\log p(\boldsymbol{x})$:

Model	**Objective**	**Key idea**
VAE	Maximise ELBO	Variational inference, 1 latent level
DDPM	Maximise ELBO (T levels)	Fixed encoder, noise prediction
GAN	**No likelihood bound**	**Adversarial game**

Fundamental difference

GANs never evaluate $\log p_{\boldsymbol{\theta}}(\boldsymbol{x})$. A **discriminator** D provides the training signal by classifying real vs. fake, driving the **generator** G to match p_r — without any likelihood computation.

Benefit: sharper samples, no ℓ_2-blurring. *Cost*: no likelihood, mode collapse risk, instability.

Generative Adversarial Networks

A **GAN** (Goodfellow et al., NeurIPS 2014) pits two neural networks against each other in a zero-sum game:

Generator G

- Input: noise $\boldsymbol{z} \sim p_z(\boldsymbol{z})$
- Output: synthetic sample $\boldsymbol{x} = G(\boldsymbol{z}; \boldsymbol{\theta}^{(g)})$
- Goal: fool the discriminator
- Implicitly defines distribution p_g

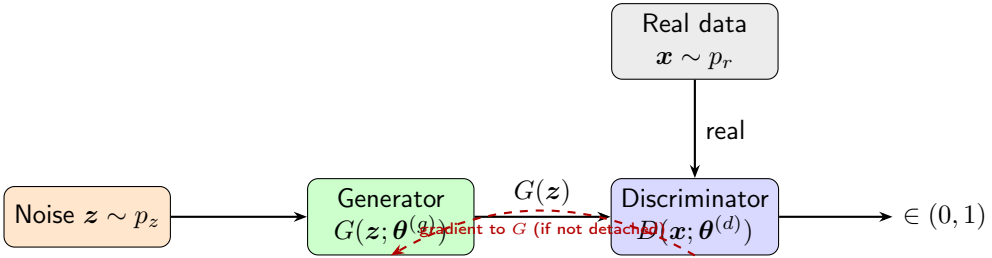
Discriminator D

- Input: image $\boldsymbol{x}$
- Output: $D(\boldsymbol{x}; \boldsymbol{\theta}^{(d)}) \in (0, 1)$
  $= \Pr(\boldsymbol{x} \text{ is real})$
- Goal: distinguish real from fake

Key insight

Neither network sees the other's parameters. They improve *each other* through adversarial feedback alone.

GAN Architecture (Schematic)



- G maps $\boldsymbol{z} \in \mathbb{R}^k \to \mathbb{R}^d$ (noise $\to$ data space)
- D maps $\boldsymbol{x} \in \mathbb{R}^d \to (0, 1)$ (image $\to$ probability)
- Training alternates: freeze G and update D; freeze D and update G

Learning Paradigms and Applications

Three learning settings:

1. **Unsupervised** — no labels; G learns the full data manifold
2. **Supervised** — conditional GAN conditions G on class label
3. **Semi-supervised** — D doubles as a feature extractor

Applications:

- Image synthesis / super-resolution
- Text-to-image translation
- Video prediction
- Data augmentation for rare events
- Anomaly detection
- Scientific simulation: particle physics, drug discovery, climate modelling

Notation and Probability Spaces

Symbol	Meaning
$p_r(\boldsymbol{x})$	Real data distribution (unknown; represented by training set)
$p_z(\boldsymbol{z})$	Latent noise prior, e.g. $\mathcal{N}(\mathbf{0}, I)$
$G(\boldsymbol{z}; \boldsymbol{\theta}^{(g)})$	Generator: differentiable map $\mathbb{R}^k \to \mathbb{R}^d$
p_g	Generator distribution = push-forward of p_z through G
$D(\boldsymbol{x}; \boldsymbol{\theta}^{(d)}) \in (0,1)$	Discriminator: $\Pr(\boldsymbol{x}$ is real$)$

Implicit distribution. For invertible G the change-of-variables gives:

$$p_g(\boldsymbol{x}) = p_z\big(G^{-1}(\boldsymbol{x})\big) \left| \det \frac{\partial G^{-1}}{\partial \boldsymbol{x}} \right|.$$

In practice G is not invertible; p_g is an *implicit* distribution accessed only through sampling.

The Minimax Objective

Value function (Goodfellow et al., 2014)

$$\min_G \max_D V(G,D) = \underbrace{\mathbb{E}_{\boldsymbol{x}\sim p_r}[\log D(\boldsymbol{x})]}_{D \text{ correct on real}} + \underbrace{\mathbb{E}_{\boldsymbol{z}\sim p_z}[\log(1-D(G(\boldsymbol{z})))]}_{D \text{ correct on fake}}$$

Discriminator (maximise V):

- $D(\boldsymbol{x}) \to 1$ on real data
- $D(G(\boldsymbol{z})) \to 0$ on generated data

Generator (minimise V):

- $D(G(\boldsymbol{z})) \to 1$ (fool D)
- $\mathbb{E}_{p_r}[\log D(\boldsymbol{x})]$ is constant w.r.t. G, so G minimises $\mathbb{E}[\log(1-D(G(\boldsymbol{z})))]$

Non-Saturating Generator Loss

Problem with the original objective. Early in training D is strong, so $D(G(\boldsymbol{z})) \approx 0$:

$$\frac{\partial}{\partial \boldsymbol{\theta}^{(g)}} \log(1 - D(G(\boldsymbol{z}))) \approx \frac{\partial}{\partial \boldsymbol{\theta}^{(g)}} \log 1 = 0.$$

The gradient *vanishes* before G has learned anything.

Fix: non-saturating loss

Instead of minimising $\log(1 - D(G(\boldsymbol{z})))$, maximise $\log D(G(\boldsymbol{z}))$:

$$\mathcal{L}_G^{\text{non-sat}} = -\mathbb{E}_{\boldsymbol{z} \sim p_z}[\log D(G(\boldsymbol{z}))].$$

Same Nash equilibrium; stronger gradient when $D(G(\boldsymbol{z})) \approx 0$.

Loss form	Value at $D(G(\boldsymbol{z})) = \varepsilon \approx 0$	|gradient|
$\log(1 - \varepsilon)$	$\approx -\varepsilon$	≈ 1 (small)
$-\log \varepsilon$	$+\infty$	$1/\varepsilon \to \infty$ (large)

The Alternating Gradient Algorithm

One training iteration

1. **Freeze** G; gradient **ascent** on D:

$$\boldsymbol{\theta}^{(d)} \leftarrow \boldsymbol{\theta}^{(d)} + \eta \nabla_{\boldsymbol{\theta}^{(d)}} V(G, D).$$

2. **Freeze** D; gradient **descent** on G:

$$\boldsymbol{\theta}^{(g)} \leftarrow \boldsymbol{\theta}^{(g)} - \eta \nabla_{\boldsymbol{\theta}^{(g)}} \mathcal{L}_G.$$

- One D step per G step in practice ($k = 1$).
- Fake images are **detached** from the computation graph during the D step to prevent spurious gradients flowing into G.
- Both networks update simultaneously — this does *not* guarantee convergence).

Deriving the Optimal Discriminator

Fix G (hence fix p_g). Write V as an integral:

$$V(G,D) = \int_{\boldsymbol{x}} \left[p_r(\boldsymbol{x}) \log D(\boldsymbol{x}) + p_g(\boldsymbol{x}) \log(1 - D(\boldsymbol{x})) \right] d\boldsymbol{x}.$$

For each $\boldsymbol{x}$ independently, maximise the integrand $f(t) = A \log t + B \log(1-t)$, $A = p_r(\boldsymbol{x})$, $B = p_g(\boldsymbol{x})$, $t = D(\boldsymbol{x}) \in (0,1)$:

$$f'(t) = \frac{A}{t} - \frac{B}{1-t} = \frac{A - (A+B)t}{t(1-t)} = 0 \implies t^* = \frac{A}{A+B}.$$

Check: $f''(t^*) = -A/(t^*)^2 - B/(1-t^*)^2 < 0$ ✓ (maximum).

Optimal discriminator

$$\boxed{D^*(\boldsymbol{x}) = \frac{p_r(\boldsymbol{x})}{p_r(\boldsymbol{x}) + p_g(\boldsymbol{x})}.}$$

This is the **Bayes-optimal classifier** for the two-class problem {real, fake} under equal class priors.

Nash Equilibrium and Loss at the Optimum

Nash equilibrium. At $p_g = p_r$:

$$D^*(\boldsymbol{x}) = \frac{p_r(\boldsymbol{x})}{p_r(\boldsymbol{x}) + p_r(\boldsymbol{x})} = \frac{1}{2}.$$

The discriminator cannot distinguish real from fake — it is reduced to random guessing.

Loss at the equilibrium. Substituting $D^* = 1/2$:

$$\begin{aligned} V(G^*, D^*) &= \log\tfrac{1}{2} \int p_r \, d\boldsymbol{x} + \log\tfrac{1}{2} \int p_r \, d\boldsymbol{x} \\ &= -2\log 2 \approx -1.386. \end{aligned}$$

Convergence indicator

In practice, monitor $D(\boldsymbol{x})$ and $D(G(\boldsymbol{z}))$ during training. Both should converge to 0.5 at equilibrium. If $D(G(\boldsymbol{z})) \ll 0.5$, the generator is losing. If $D(\boldsymbol{x}) \ll 0.5$, the discriminator is failing.

KL and JS Divergences

Kullback-Leibler divergence:

$$D_{\mathrm{KL}}(p\|q) = \int p(\boldsymbol{x}) \log \frac{p(\boldsymbol{x})}{q(\boldsymbol{x})}\, d\boldsymbol{x} \;\geq 0,$$

with equality iff $p = q$ a.e. (Gibbs' inequality). Note: $D_{\mathrm{KL}}(p\|q) \neq D_{\mathrm{KL}}(q\|p)$ in general.

Jensen-Shannon divergence (symmetric, bounded):

$$D_{\mathrm{JS}}(p\|q) = \frac{1}{2} D_{\mathrm{KL}}\left(p \,\middle\|\, \frac{p+q}{2}\right) + \frac{1}{2} D_{\mathrm{KL}}\left(q \,\middle\|\, \frac{p+q}{2}\right).$$

Properties of D_{JS}

- Symmetric: $D_{\mathrm{JS}}(p\|q) = D_{\mathrm{JS}}(q\|p)$
- Non-negative: $D_{\mathrm{JS}}(p\|q) \geq 0$, with equality iff $p = q$
- Bounded: $D_{\mathrm{JS}}(p\|q) \in [0, \log 2]$ for any p, q
- Related to total variation: $D_{\mathrm{JS}}(p\|q) \leq \log 2 \cdot \mathrm{TV}(p, q)$

GAN Loss = Jensen-Shannon Divergence

Expanding $D_{\mathrm{JS}}(p_r \| p_g)$ and simplifying:

$$D_{\mathrm{JS}}(p_r \| p_g) = \frac{1}{2} D_{\mathrm{KL}}\left(p_r \,\middle\|\, \frac{p_r + p_g}{2}\right) + \frac{1}{2} D_{\mathrm{KL}}\left(p_g \,\middle\|\, \frac{p_r + p_g}{2}\right)$$
$$= \frac{1}{2}\left(\log 2 + \int p_r \log \frac{p_r}{p_r + p_g}\, d\boldsymbol{x}\right) + \frac{1}{2}\left(\log 2 + \int p_g \log \frac{p_g}{p_r + p_g}\, d\boldsymbol{x}\right).$$

Recognising $V(G, D^*)$ in the integrals:

Main theorem (Goodfellow et al. 2014)

$$\boxed{V(G, D^*) = 2\, D_{\mathrm{JS}}(p_r \| p_g) - 2 \log 2.}$$

Conclusion. The GAN minimax objective is equivalent to minimising the Jensen-Shannon divergence between p_r and p_g. The global minimum $-2 \log 2$ is achieved when $p_g = p_r$.

Binary Cross-Entropy Losses

Discriminator loss (real label $= s$, fake label $= 0$):

$$\mathcal{L}_D = -\frac{1}{N}\sum_{i=1}^{N}\log D(\boldsymbol{x}_i) - \frac{1}{M}\sum_{j=1}^{M}\log\bigl(1 - D(G(\boldsymbol{z}_j))\bigr).$$

Generator loss (non-saturating):

$$\mathcal{L}_G = -\frac{1}{M}\sum_{j=1}^{M}\log D(G(\boldsymbol{z}_j)).$$

One-sided label smoothing ($s = 0.9$)

Replace real target 1 with $s < 1$. The optimal discriminator becomes $D_s^*(\boldsymbol{x}) = s \cdot p_r(\boldsymbol{x})/(p_r(\boldsymbol{x}) + p_g(\boldsymbol{x}))$, preventing overconfidence and keeping gradients to G alive.

Wasserstein Distance (WGAN)

Problem with JS divergence. When p_r and p_g have non-overlapping supports (common early in training): $D_{\mathrm{JS}}(p_r \| p_g) = \log 2$ constantly, giving zero gradient everywhere.

Wasserstein-1 distance (*Earth-mover*):

$$W(p_r, p_g) = \inf_{\gamma \in \Pi(p_r, p_g)} \mathbb{E}_{(\boldsymbol{x}, \mathbf{y}) \sim \gamma} \|\boldsymbol{x} - \mathbf{y}\|$$

$\Pi(p_r, p_g)$: all couplings with marginals p_r, p_g.

Kantorovich-Rubinstein duality

$$W(p_r, p_g) = \sup_{\|f\|_L \leq 1} \big(\mathbb{E}_{p_r}[f] - \mathbb{E}_{p_g}[f]\big).$$

Sup. over all 1-Lipschitz functions f.

Advantage

W gives meaningful gradients even when $p_r \cap p_g = \emptyset$. Training is more stable.

WGAN: Gradient Penalty

WGAN critic. Replace D with a critic f_w (no sigmoid):

$$\mathcal{L}_{\text{critic}} = \mathbb{E}_{p_g}[f_w(\boldsymbol{x})] - \mathbb{E}_{p_r}[f_w(\boldsymbol{x})] + \lambda\,\mathcal{L}_{\text{GP}}.$$

Gradient penalty (Gulrajani et al., 2017):

$$\mathcal{L}_{\text{GP}} = \mathbb{E}_{\hat{\boldsymbol{x}}}\Big[\big(\|\nabla_{\hat{\boldsymbol{x}}} f_w(\hat{\boldsymbol{x}})\|_2 - 1\big)^2\Big],$$

where $\hat{\boldsymbol{x}} = \epsilon\boldsymbol{x} + (1-\epsilon)G(\boldsymbol{z})$, $\epsilon \sim U[0,1]$ (uniform on straight lines between real and fake points).

- Enforces the 1-Lipschitz constraint via soft penalisation.
- More stable than weight clipping (original WGAN).
- $\lambda = 10$ is a standard choice.

Vanishing Gradient

When does it occur? When D is perfect early in training: $D(\boldsymbol{x}) = 1 \; \forall \boldsymbol{x} \sim p_r$ and $D(G(\boldsymbol{z})) = 0 \; \forall \boldsymbol{z} \sim p_z$.

Original G gradient:

$$\nabla_{\boldsymbol{\theta}^{(g)}} \mathbb{E}[\log(1 - D(G(\boldsymbol{z})))] \approx \nabla_{\boldsymbol{\theta}^{(g)}} \mathbb{E}[\log 1] = \mathbf{0}.$$

No information flows to G.

Non-saturating loss $-\log D(G(\boldsymbol{z}))$:

$$\nabla_{\boldsymbol{\theta}^{(g)}} \big(-\log D(G(\boldsymbol{z}))\big) = -\frac{1}{D(G(\boldsymbol{z}))} \nabla_{\boldsymbol{\theta}^{(g)}} D(G(\boldsymbol{z})).$$

When $D(G(\boldsymbol{z})) \approx 0$, the factor $1/D(G(\boldsymbol{z})) \to \infty$ amplifies the gradient — learning continues.

Mode Collapse

Definition. The generator maps many different noise vectors z to the *same* (or very few) output(s):

$$\mathrm{supp}(p_g) \ll \mathrm{supp}(p_r).$$

The JS divergence can still be low on the covered modes, so the loss does not detect collapse.

Why it happens. Given the current discriminator, G finds a single "safe" output that fools D; D then adapts, and G shifts to another single output. The two networks chase each other indefinitely.

Mitigations:

- **Minibatch discrimination**: D sees statistics of a batch, not individual images — diversity is rewarded.
- **Unrolled GANs**: G optimises against a future D, preventing short-sighted collapse.
- **Spectral normalisation**: stabilises D by bounding its Lipschitz constant.
- **WGAN**: Wasserstein distance measures full distribution mismatch, not just modes.

Non-Convergence and the Game-Theoretic View

GAN training solves a **two-player non-cooperative game**:

$$G^* = \arg\min_G \arg\max_D V(G, D).$$

Nash equilibrium exists at $p_g = p_r$, $D^* \equiv \frac{1}{2}$, but:

- Simultaneous gradient updates do *not* guarantee convergence to a Nash equilibrium.
- The loss landscape of each player changes at every step (non-stationary environment).
- If $\max_D V(G, D)$ is not convex in $\boldsymbol{\theta}^{(g)}$, global convergence fails even in theory.

Practical stabilisation:

- Lower β_1 in Adam (reduces momentum: 0.5 instead of 0.9)
- One-sided label smoothing
- Spectral normalisation of D's weights
- Progressive growing (ProGAN), self-attention (SAGAN)
- Two-time-scale updates: different learning rates for G and D

Evaluation Metrics: Overview

Evaluating generative models is hard: there is no ground truth output to compare against.

Metric	Measures	Limitation
Inception Score (IS)	Quality + diversity of generated images	Requires pretrained classifier
FID	Distance between real/fake feature distributions	Sensitive to sample size
MMD	Kernel-based distribution distance	Choice of kernel matters
Precision/Recall	Coverage vs. fidelity separately	Requires feature extractor

Maximum Mean Discrepancy (MMD)

Definition in RKHS. Let $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ be a kernel. The **MMD** between distributions p, q is:

$$\mathrm{MMD}^2(p, q) = \mathbb{E}_{\boldsymbol{x},\boldsymbol{x}'\sim p}[k(\boldsymbol{x}, \boldsymbol{x}')] - 2\,\mathbb{E}_{\boldsymbol{x}\sim p,\,\mathbf{y}\sim q}[k(\boldsymbol{x}, \mathbf{y})] + \mathbb{E}_{\mathbf{y},\mathbf{y}'\sim q}[k(\mathbf{y}, \mathbf{y}')].$$

With the Gaussian kernel $k(\boldsymbol{x}, \mathbf{y}) = \exp(-\|\boldsymbol{x} - \mathbf{y}\|^2/2\sigma^2)$: $\mathrm{MMD} = 0$ iff $p = q$ (characteristic kernel property).

Unbiased empirical estimator

Given n real samples $\{\boldsymbol{x}_i\}$ and m fake samples $\{\mathbf{y}_j\}$:

$$\widehat{\mathrm{MMD}}^2 = \frac{1}{n(n-1)}\sum_{i\neq j} k(\boldsymbol{x}_i, \boldsymbol{x}_j) - \frac{2}{nm}\sum_{i,j} k(\boldsymbol{x}_i, \mathbf{y}_j) + \frac{1}{m(m-1)}\sum_{i\neq j} k(\mathbf{y}_i, \mathbf{y}_j).$$

Diagonal terms ($i = j$) excluded for unbiasedness.

Fréchet Inception Distance (FID)

FID is currently the most widely used GAN metric.

Procedure:

1. Extract feature vectors from the **InceptionV3** penultimate layer for n real images and n generated images.
2. Fit Gaussians: $\mathcal{N}(\mu_r, \Sigma_r)$ and $\mathcal{N}(\mu_g, \Sigma_g)$.
3. Compute the **Fréchet distance**:

$$\mathrm{FID} = \|\mu_r - \mu_g\|^2 + \mathrm{Tr}\Big(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\Big).$$

- Lower FID = better quality and diversity.
- State-of-the-art GANs achieve FID < 5 on MNIST; < 10 on CIFAR-10.
- Sensitive to sample size n — use $\geq 10\,000$ samples.

Generator Architecture for MNIST

Fully-connected generator (Goodfellow et al., 2014 style):

$$z \in \mathbb{R}^{100} \xrightarrow{\text{Lin+BN+LReLU}} \mathbb{R}^{256} \xrightarrow{\text{Lin+BN+LReLU}} \mathbb{R}^{512} \xrightarrow{\text{Lin+BN+LReLU}} \mathbb{R}^{1024} \xrightarrow{\text{Lin+Tanh}} \mathbb{R}^{784}$$

Design choices:

- **BatchNorm** after each Linear layer: normalises activations, allows higher LR, stabilises gradient flow
- **LeakyReLU(0.2):** prevents dying ReLU; small gradient for negative inputs
- **Tanh** output: maps to $[-1, 1]$, matching MNIST normalised to $[-1, 1]$
- Weights: $\mathcal{N}(0, 0.02^2)$ (DCGAN convention)

Discriminator differences:

- **No BatchNorm**: introduces minibatch dependencies, causes instability in D
- **Dropout(0.3)**: regularises, prevents D from over-fitting on real data
- **Sigmoid** output: probability in $(0, 1)$
- Input: flattened image $\in \mathbb{R}^{784}$

Latent Space Interpolation

A well-trained generator has a **smooth latent space**. Linear interpolation between two noise vectors:

$$\boldsymbol{z}(\alpha) = (1-\alpha)\,\boldsymbol{z}_1 + \alpha\,\boldsymbol{z}_2, \quad \alpha \in [0,1]$$

should yield a smooth visual transition between the corresponding images.

- **Smooth transitions**: generator has learned a good manifold.
- **Abrupt jumps**: mode collapse or poor latent coverage.
- Provides a qualitative test of latent space geometry that FID cannot capture.

Spherical interpolation

On a high-dimensional sphere (approximately the case for $\boldsymbol{z} \sim \mathcal{N}(0, I)$), *spherical* interpolation $\boldsymbol{z}(\alpha) = \frac{\sin((1-\alpha)\Omega)}{\sin\Omega}\boldsymbol{z}_1 + \frac{\sin(\alpha\Omega)}{\sin\Omega}\boldsymbol{z}_2$, $\cos\Omega = \hat{\boldsymbol{z}}_1 \cdot \hat{\boldsymbol{z}}_2$, stays on the noise manifold and gives more natural transitions.

Conditional GAN (cGAN)

A **conditional GAN** (Mirza & Osindero, 2014) conditions both G and D on auxiliary information $\mathbf{y}$ (class label, image, text):

$$\min_G \max_D V(G,D) = \mathbb{E}[\log D(\boldsymbol{x}|\mathbf{y})] + \mathbb{E}[\log(1 - D(G(\boldsymbol{z}|\mathbf{y})))].$$

For MNIST: $\mathbf{y}$ is the digit class (one-hot, $\mathbb{R}^{10}$). Concatenate $\mathbf{y}$ to both G's input $\boldsymbol{z}$ and D's input $\boldsymbol{x}$.

cGAN enables

- Controlled generation: specify which digit to generate
- Image-to-image translation (pix2pix)
- Text-to-image synthesis
- Style transfer

Deep Convolutional GAN (DCGAN)

Radford, Metz, Chintala (ICLR 2016) replaced fully-connected layers with **strided convolutions**:

- **Generator**: fractionally strided convolutions (transposed conv) to upsample from $\boldsymbol{z} \in \mathbb{R}^{100}$ to full-resolution image.
- **Discriminator**: strided convolutions to downsample.
- BatchNorm in G (all layers except output); BatchNorm in D (except input).
- ReLU in G; LeakyReLU in D.
- No pooling layers; no fully-connected layers (except for $\boldsymbol{z}$).

Key advance

DCGAN produced the first visually convincing high-resolution GAN outputs and established the architectural conventions still used today. The DCGAN paper's guidelines (no BN in D input layer, LReLU slope 0.2, Adam with $\beta_1 = 0.5$) remain standard practice.

Comparison of GAN Variants

Variant	**Key change**	**Loss**	**Advantage**
Vanilla GAN	FC networks	JS div.	Simple
DCGAN	Conv. nets	JS div.	Better images
WGAN	Wt. clipping	Wasserstein	More stable
WGAN-GP	Grad. penalty	Wasser.+GP	No clipping
cGAN	Conditioning	JS or W	Controlled
StyleGAN	Style-based	W	SOTA

Key Equations: Summary

Concept	Formula
Minimax objective	$\min_G \max_D \ \mathbb{E}_{p_r}[\log D(\boldsymbol{x})] + \mathbb{E}_{p_z}[\log(1 - D(G(\boldsymbol{z})))]$
Optimal discriminator	$D^*(\boldsymbol{x}) = p_r(\boldsymbol{x})/(p_r(\boldsymbol{x}) + p_g(\boldsymbol{x}))$
Nash equilibrium	$p_g = p_r, \quad D^* \equiv 1/2$
Loss at equilibrium	$V(G^*, D^*) = -2\log 2$
GAN loss = JS div.	$V(G, D^*) = 2D_{\mathrm{JS}}(p_r \Vert p_g) - 2\log 2$
Non-sat. generator	$\mathcal{L}_G = -\mathbb{E}_{p_z}[\log D(G(\boldsymbol{z}))]$
Discriminator loss	$\mathcal{L}_D = -\mathbb{E}_{p_r}[\log D(\boldsymbol{x})] - \mathbb{E}_{p_z}[\log(1 - D(G(\boldsymbol{z})))]$
Wasserstein distance	$W(p_r, p_g) = \sup_{\Vert f \Vert_L \leq 1}(\mathbb{E}_{p_r}[f] - \mathbb{E}_{p_g}[f])$
Gradient penalty	$\lambda\, \mathbb{E}[(\Vert \nabla f_w(\hat{\boldsymbol{x}}) \Vert_2 - 1)^2]$
MMD	$\sqrt{\mathbb{E}_{p\times p}[k] - 2\mathbb{E}_{p\times q}[k] + \mathbb{E}_{q\times q}[k]}$
FID	$\Vert \mu_r - \mu_g \Vert^2 + \mathrm{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2})$

Exercises

1. **Optimal discriminator.** Derive $D^*(\boldsymbol{x}) = p_r/(p_r + p_g)$ by differentiating $f(t) = A\log t + B\log(1-t)$ and verify it is a maximum, not a minimum.
2. **JS divergence bounds.** Prove $D_{\mathrm{JS}}(p\|q) \in [0, \log 2]$ for any two distributions and find the conditions for equality on each side.
3. **Vanishing gradient.** At $D(G(\boldsymbol{z})) = \varepsilon \ll 1$, compute $|\nabla_{\boldsymbol{\theta}^{(g)}} \mathcal{L}_G^{\text{orig}}|$ and $|\nabla_{\boldsymbol{\theta}^{(g)}} \mathcal{L}_G^{\text{non-sat}}|$ and compare their magnitudes as $\varepsilon \to 0$.
4. **Label smoothing.** Derive the optimal discriminator $D_s^*(\boldsymbol{x})$ when real targets are smoothed to $s \in (0,1)$ instead of 1. Show it converges to D^* as $s \to 1$.
5. **Wasserstein distance.** Compute $W(\mathcal{N}(0,1), \mathcal{N}(\mu,1))$ analytically and verify that $W \to 0$ as $\mu \to 0$, unlike D_{JS} when the two Gaussians have non-overlapping support.
6. **Conditional GAN.** Extend the minimax objective to the conditional setting. What changes in the derivation of $D^*(\boldsymbol{x}|\mathbf{y})$?

Three Families: A Unified Mathematical View

All three learn $p_{\boldsymbol{\theta}}(\boldsymbol{x})$ but with different objectives and inductive biases.

Criterion	**VAE**	**Diffusion**	**GAN**
Objective	ELBO	ELBO (T terms)	Minimax / JS
Likelihood	✓	✓	×
Latent space	Compact $\boldsymbol{h}$	Full-dim $\boldsymbol{x}_t$	Noise $\boldsymbol{z}$
Encoder	Learned	Fixed schedule	None
Sampling cost	1 decoder pass	T steps	1 pass
Stability	High	High	Low–Moderate
Sample quality	Moderate	State of the art	High (sharp)
Mode coverage	Good	Excellent	Partial
Anomaly detect.	✓	✓	×

Why VAEs Produce Blurry Samples

Mathematical reason. The VAE decoder minimises:

$$-\mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{h}|\boldsymbol{x})}\big[\log p_{\boldsymbol{\theta}}(\boldsymbol{x}|\boldsymbol{h})\big] \;\propto\; \mathbb{E}_{q_{\boldsymbol{\phi}}}\big[\|\boldsymbol{x} - \hat{\boldsymbol{x}}_{\boldsymbol{\theta}}(\boldsymbol{h})\|^2\big].$$

Because the posterior $q_{\boldsymbol{\phi}}(\boldsymbol{h}|\boldsymbol{x})$ has nonzero variance, the decoder learns the **posterior mean over all likely completions**, which is a blur over the data manifold.

Contrast with the other two models:

Diffusion avoids blurring by

predicting only the *noise added at one step* t, not the full image. Each reverse step makes a small, committed correction. The iterative nature removes the need to average over the full distribution of plausible outputs.

GANs avoid blurring by

replacing the fixed ℓ_2 loss with an *adaptive, learned* discriminator loss. The discriminator penalises blurriness and any systematic deviation from real data, providing a richer training signal than any fixed pixel-wise objective.

The Likelihood Problem in GANs

A fundamental asymmetry. VAEs and diffusion models optimise a lower bound on $\log p(\boldsymbol{x})$, so they can both *generate* and *evaluate* density. GANs have **no access to** $p_g(\boldsymbol{x})$.

Consequences of no likelihood:

- Cannot compute log-likelihood scores for anomaly detection.
- Cannot evaluate model fit on held-out data in a principled way.
- Evaluation requires proxy metrics (FID, IS, MMD) rather than test-set log-likelihood.
- Cannot be embedded as a prior in a Bayesian framework without approximation.

Why this is a price worth paying (sometimes). The discriminator provides a *perceptual* loss that no fixed $p_{\boldsymbol{\theta}}$ objective can replicate: it adapts to what "looks real" to a neural network, capturing texture, sharpness, and high-frequency detail that ℓ_2 and even ELBO objectives systematically underweight.

Training Stability: A Comparison

VAE — Stable

ELBO is a well-defined scalar. Both networks trained with standard SGD. KL prevents collapse when weighted correctly. Main challenge: choosing β for β-VAE.

Diffusion — Stable

MSE on noise prediction is smooth. No adversarial dynamics. Low MC gradient variance ($\boldsymbol{x}_t$ sampled in one step). Main challenge: noise schedule design.

GAN — Unstable

Non-stationary minimax landscape. No Nash convergence guarantee. $V(G, D)$ is concave-convex; simultaneous gradient ascent-descent diverges even for bilinear games. Requires WGAN-GP, spectral norm, label smoothing.

Key insight

GAN instability is *fundamental*: the minimax saddle structure prevents convergence guarantees that both VAE and DDPM enjoy.

Mode Collapse vs. Posterior Averaging: Two Failure Modes

The two pathological failure modes of generative models lie at opposite ends:

Posterior averaging (VAE)

Too much spread. The decoder must explain all data through a single mean. Result: generated images are averages over the data manifold — blurry. The model *covers* all modes but fails to commit to any one of them.

$$\hat{\boldsymbol{x}} = \mathbb{E}_{q_{\boldsymbol{\phi}}(\boldsymbol{h}|\boldsymbol{x})}[\hat{\boldsymbol{x}}_{\boldsymbol{\theta}}(\boldsymbol{h})] \;\approx\; \text{mean of modes}$$

Mode collapse (GAN)

Too little spread. The generator finds a single safe output that fools D and ignores the rest of p_r. Result: sharp images but only from a few modes. $\mathrm{supp}(p_g) \ll \mathrm{supp}(p_r)$.

$$G(\boldsymbol{z}) \approx \boldsymbol{x}^* \;\forall \boldsymbol{z} \;\Rightarrow\; D_{\mathrm{JS}}(p_r \| p_g) \approx \log 2 \text{ (flat)}$$

Diffusion avoids both: the fixed encoder $q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})$ prevents collapse (no network can "cheat" the noise schedule), and the iterative denoising avoids averaging because each step commits to only a small noise correction rather than a full image prediction.

Detailed Pros and Cons: VAEs

Pros

- **Likelihood bound**: ELBO usable for anomaly detection.
- **Compact latent**: $\boldsymbol{h}$ supports interpolation, clustering, disentanglement, property prediction.
- **Fast generation**: single decoder pass.
- **Stable training**: well-defined ELBO, no adversary.
- **Flexible**: β-VAE for disentanglement; hierarchical VAEs for expressivity.
- **Scientific use**: ideal for molecules, spectra, simulations.

Cons

- **Blurry samples**: posterior averaging is fundamental, not a capacity issue.
- **Posterior collapse**: decoder ignores $\boldsymbol{h}$; KL$\to 0$, latent useless.
- **ELBO gap**: $\log p(\boldsymbol{x}) = \mathrm{ELBO} + D_{\mathrm{KL}}(q_{\boldsymbol{\phi}} \| p(\boldsymbol{h}|\boldsymbol{x})) > \mathrm{ELBO}$.
- **Mean-field**: factorised Gaussian misses correlated posteriors.
- **Fixed-dim bottleneck**: wrong d_h hurts quality.

Detailed Pros and Cons: Diffusion Models

Pros

- **Best sample quality**: sharp, diverse, no blurring.
- **Full coverage**: fixed forward process prevents mode collapse.
- **Stable training**: MSE on noise, no adversarial dynamics.
- **Likelihood bound**: ELBO $\rightarrow \log p(\boldsymbol{x})$ as $T \rightarrow \infty$.
- **Flexible conditioning**: text, class, depth via classifier-free guidance and cross-attention.
- **Rich theory**: SDE/score connection (Song et al. 2021).

Cons

- **Slow sampling**: $T = 100$–1000 steps (DDIM/DPM-Solver: 10–50).
- **No compact latent**: full-dim $\boldsymbol{x}_t$ precludes clustering or property prediction.
- **Compute-intensive**: large T, high-dim $\boldsymbol{x}$ need significant GPU memory.
- **Poor for repr. learning**: no encoder for features.
- **Schedule-sensitive**: cosine preferred over linear at high resolution.

Detailed Pros and Cons: GANs

Pros

- **Sharpest samples**: adversarial loss catches blurriness that ℓ_2/ELBO miss.
- **Fastest inference**: single generator pass.
- **Flexible loss**: works when likelihood is intractable.
- **Rich ecosystem**: cGAN, DCGAN, WGAN-GP, StyleGAN, pix2pix.
- **Image translation**: handles paired and unpaired transfer.

Cons

- **No likelihood**: no anomaly detection or density estimation.
- **Mode collapse**: G covers only part of p_r; FID/IS may not detect it.
- **Instability**: non-stationary minimax; no convergence guarantee; hyperparameter-sensitive.
- **JS flatness**: $D_{\text{JS}} = \log 2$ when supports don't overlap (WGAN fixes this).
- **Wasted discriminator**: D discarded after training.

When to Use Each Model

Use case	Model	Reason
Image / audio / video synthesis	Diffusion	Best quality, no collapse
Text-to-image (Stable Diffusion)	LDM	VAE compression + diffusion
Image-to-image translation	GAN	Adversarial perceptual loss
Real-time generation	VAE or GAN	Single-pass inference
Anomaly detection	VAE or Diffusion	Likelihood bound available
Representation / clustering	VAE	Compact structured latent
Scientific data (molecules)	VAE	Low-dim latent for prediction
Data augmentation	GAN or Diffusion	Sharp, diverse samples
Inpainting / super-resolution	Diffusion	Score-based posterior
Very limited data	VAE	Stable; KL regularisation

Key Equations: Complete Summary Across All Three Models

Concept	VAE / Diffusion	GAN
Encoder	$q_{\boldsymbol{\phi}}(\boldsymbol{h}\|\boldsymbol{x}) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\phi}}, \boldsymbol{\sigma}^2_{\boldsymbol{\phi}}\mathbf{I})$	None (implicit via G)
Marginal / forward	$q(\boldsymbol{x}_t\|\boldsymbol{x}_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\boldsymbol{x}_{t-1}, (1-\alpha_t)\mathbf{I})$	$\boldsymbol{x} = G(\boldsymbol{z}; \boldsymbol{\theta}^{(g)})$
Objective	$\mathcal{L} = \mathbb{E}[\log p_{\boldsymbol{\theta}}(\boldsymbol{x}\|\boldsymbol{h})] - D_{\mathrm{KL}}(q\|p)$	$\min_G \max_D \mathbb{E}[\log D(\boldsymbol{x})] + \mathbb{E}[\log(1 - D(G(\boldsymbol{z})))]$
Optimal solution	DDPM: $\boldsymbol{\mu}_{\boldsymbol{\theta}} = \frac{1}{\sqrt{\alpha_t}}(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_{\boldsymbol{\theta}})$	$D^*(\boldsymbol{x}) = \frac{p_r(\boldsymbol{x})}{p_r(\boldsymbol{x}) + p_g(\boldsymbol{x})}$
Loss at equilibrium	VAE ELBO $\to \log p(\boldsymbol{x})$	$V(G^*, D^*) = -2\log 2$
Divergence minimised	$D_{\mathrm{KL}}(q\|p)$ at each level	$D_{\mathrm{JS}}(p_r\|p_g)$
Score connection	$\boldsymbol{s}_{\boldsymbol{\theta}} = -\boldsymbol{\epsilon}_{\boldsymbol{\theta}}/\sqrt{1-\bar{\alpha}_t}$	$D^* \to$ implicit score estimator
Sampling	1 pass (VAE); T steps (DDPM)	1 pass

Exercises: Cross-Model Comparisons

Exercises 1–3

1. **Common ELBO.** Show VAE (1 level) and DDPM (T levels) are special cases of the MHVAE ELBO. Identify encoder, decoder, prior in each.
2. **Blurriness from ℓ_2.** For $p_r = \frac{1}{2}\delta(x-a) + \frac{1}{2}\delta(x-b)$, show the VAE decoder learns $\hat{x} = (a+b)/2$ while a GAN can produce both modes.
3. **Convergence.** State which property of the DDPM ELBO makes it amenable to SGD while the GAN minimax saddle is not.

Exercises 4–6

4. **Wasserstein vs. JS.** Compute $D_{\mathrm{JS}}(\mathcal{N}(0,1)\|\mathcal{N}(\mu,1))$ and $W(\mathcal{N}(0,1),\mathcal{N}(\mu,1))$. Show D_{JS} saturates at $\log 2$ while $W = |\mu|$ grows.
5. **Score matching bridge.** Using $\nabla_{\boldsymbol{x}_t} \log q(\boldsymbol{x}_t|\boldsymbol{x}_0) = -\boldsymbol{\epsilon}/\sqrt{1-\bar{\alpha}_t}$, show DDPM equals denoising score matching (Vincent 2011).
6. **Latent Diffusion.** Express the LDM objective as a composition of VAE ELBO and DDPM ELBO. Identify which parameters are trained in each stage.

References

Diffusion models

1. Kingma & Welling, *Auto-Encoding Variational Bayes*, ICLR 2014. 1312.6114
2. Ho, Jain & Abbeel, *DDPM*, NeurIPS 2020. 2006.11239
3. Song et al., *Score-Based Generative Modeling through SDEs*, ICLR 2021. 2011.13456
4. Song, Meng & Ermon, *DDIM*, ICLR 2021. 2010.02502
5. Kingma et al., *Variational Diffusion Models*, NeurIPS 2021. 2107.00630
6. Nichol & Dhariwal, *Improved DDPM*, ICML 2021. 2102.09672
7. Rombach et al., *Latent Diffusion Models*, CVPR 2022. 2112.10752
8. Luo, *Understanding Diffusion Models*, 2022. 2208.11970

Generative Adversarial Networks

1. Goodfellow et al., *Generative Adversarial Nets*, NeurIPS 2014. 1406.2661
2. Mirza & Osindero, *Conditional GAN*, 2014. 1411.1784
3. Radford, Metz & Chintala, *DCGAN*, ICLR 2016. 1511.06434
4. Arjovsky, Chintala & Bottou, *WGAN*, ICML 2017. 1701.07875
5. Gulrajani et al., *WGAN-GP*, NeurIPS 2017. 1704.00028
6. Heusel et al., *FID*, NeurIPS 2017. 1706.08500
7. Goodfellow, Bengio & Courville, *Deep Learning*, MIT Press 2016, Ch. 20.
8. Weng, *From GAN to WGAN*, blog post, 2017.

Course at a Glance

Two-Part Structure

1. Part I: Discriminative Models
   Jan–Mar: NNs, CNNs, RNNs, autoencoders, transformers
2. Part II: Generative Models
   Apr–May: Boltzmann machines, VAEs, diffusion, GANs

Methods Covered

Discriminative

- Feed-forward NNs
- PINNs
- CNNs
- RNNs / LSTMs
- Autoencoders
- Transformers

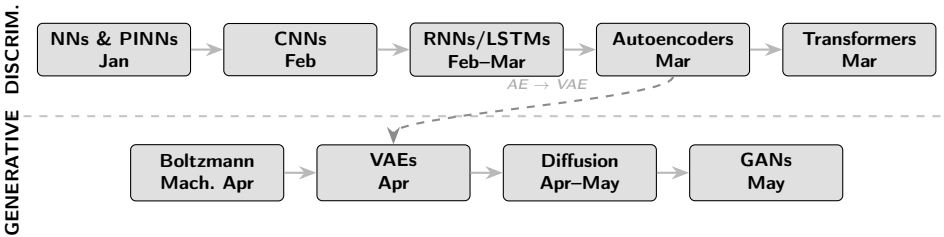
Generative

- Boltzmann machines
- VAEs
- Diffusion models
- GANs

Unifying Thread

Deep learning mathematics applied directly to problems in the physical sciences.

Course Road Map



Part I

Discriminative Deep Learning

January – March

Week 1 (Jan 19–23) — Course Introduction & NN Basics

Mathematics of Deep Learning

- Neurons, layers, activation functions
- Forward pass (compact form): $\boldsymbol{a}^{(l)} = \sigma\big(\boldsymbol{W}^{(l)}\boldsymbol{a}^{(l-1)} + \boldsymbol{b}^{(l)}\big)$
- Loss functions and optimisation
- Backpropagation: chain rule
- Writing a NN from scratch

Overview & Projects

- Course structure and themes
- Deep learning in the physical sciences: survey
- Discussion of project topics
- Tools: PyTorch / JAX

Goal

Build an NN from first principles before using high-level frameworks.

Week 2 (Jan 26–30) — NNs for Differential Equations

Deep Learning Mathematics (cont.)

- Optimisers: SGD, Adam, RMSProp
- Regularisation: dropout, batch norm
- Universal approximation theorem
- Practical NN coding session

Solving ODEs / PDEs

- NNs as mesh-free function approximators
- Loss encodes boundary conditions
- Advantages and limitations
- Code examples: simple ODEs

Bridge to the Physical Sciences

NNs that solve differential equations are one of the most direct applications of deep learning to the physical sciences.

Week 3 (Feb 2–6) — Physics-Informed Neural Networks (PINNs)

PINNs: Key Idea

Embed the PDE directly into the loss function:

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + \lambda \mathcal{L}_{\text{BC}}$$

- Automatic differentiation for residuals
- Hard vs. soft BC enforcement
- Schrödinger, heat, Poisson as examples

Implementation & Results

- Network architecture choices
- Collocation point sampling
- Training strategies
- Benchmarking vs. classical solvers
- Limitations of PINNs

Weeks 4–5 (Feb 9–20) — Convolutional Neural Networks

Week 4: CNN Mathematics

- Discrete convolution and cross-correlation
- Filters, feature maps, receptive fields
- Pooling: max, average, global
- Parameter sharing, equivariance
- Backprop through conv layers

Week 5: Architectures

- Classic architectures: LeNet, ResNet, U-Net
- Skip connections and residual learning
- CNNs for physics: images, fields, spectra

Applications

CNNs excel at identifying spatial patterns in simulation output, detector images, and lattice field configurations.

Weeks 6–7 (Feb 23 – Mar 6) — Recurrent Neural Networks & LSTMs

Recurrent Neural Networks

- Sequential data: time series, trajectories
- Vanilla RNN hidden-state update:
  $\boldsymbol{h}_t = \tanh(\boldsymbol{W}_{hh}\boldsymbol{h}_{t-1} + \boldsymbol{W}_{xh}\boldsymbol{x}_t + \boldsymbol{b})$
- Backpropagation through time (BPTT)
- Vanishing / exploding gradients

Long Short-Term Memory

- LSTM gates: forget, input, output
- Cell state as long-range memory
- GRUs as a lighter alternative
- Applications:
  molecular dynamics,
  time-series forecasting

Weeks 8–9 (Mar 9–20) — Autoencoders & Dimensionality Reduction

Autoencoders

Encoder–decoder: $\boldsymbol{x} \to \boldsymbol{z} \to \hat{\boldsymbol{x}}$

- Bottleneck latent space $\boldsymbol{z}$
- Reconstruction loss $\|\boldsymbol{x} - \hat{\boldsymbol{x}}\|^2$
- Denoising autoencoders
- Convolutional AE architectures

Links to PCA & Implementations

- Linear AE $\equiv$ PCA (exactly)
- Kernel PCA, nonlinear extensions
- Latent space visualisation
- Applications: order parameters, phase classification, compression

Week 10 (Mar 23–27) — Transformers & Part I Summary

The Transformer Architecture

Self-attention:

$$\mathrm{Att}(Q,K,V) = \mathrm{softmax}\big(QK^\top/\sqrt{d_k}\big)V$$

- Multi-head attention
- Positional encoding
- Encoder / decoder stacks
- Applications: quantum chemistry, PDEs and many other

Part I Summary

Five discriminative families:

1. Feed-forward NNs & PINNs
2. CNNs
3. RNNs / LSTMs
4. Autoencoders
5. Transformers

Each tackles a distinct structural challenge in data from the physical sciences.

Part II

Generative Deep Learning

April – May

Week 11 (Apr 6–10) — Generative Models & Boltzmann Machines

What Are Generative Models?

- Goal: learn $p_\theta(\boldsymbol{x}) \approx p_{\text{data}}(\boldsymbol{x})$
- Explicit vs. implicit density estimation
- Energy-based models: $p_\theta(\boldsymbol{x}) = e^{-E_\theta(\boldsymbol{x})}/Z(\theta)$
- Markov Chain Monte Carlo
- Gibbs sampling

Boltzmann Machines

- Restricted BM: visible & hidden units
- Contrastive Divergence (CD-k)
- Persistent CD and variants
- Deep Boltzmann Machines
- Statistical mechanics analogy

Week 12 (Apr 13–17) — Energy-Based Models

Energy-Based Models (Deeper Dive)

- Score matching and noise contrastive estimation
- Contrastive Divergence revisited
- Expressivity of energy functions
- Links to partition functions in statistical physics
- Training instabilities and remedies

Boltzmann Machine Codes

- Implementing an RBM from scratch
- Sampling via Gibbs chains
- Visualising learned features
- Wave-function ansatz application
- Benchmarks and diagnostics

Highlight

RBMs have been used as variational ansätze for quantum many-body ground states — a direct link to quantum computing.

Week 13 (Apr 20–24) — Variational Autoencoders (VAEs)

VAE: Probabilistic Autoencoders

- Latent variable model:
  $p(\boldsymbol{x}) = \int p(\boldsymbol{x}|\boldsymbol{z})p(\boldsymbol{z})\,\mathrm{d}\boldsymbol{z}$
- ELBO objective:
  $\mathcal{L} = \mathbb{E}_q[\log p(\boldsymbol{x}|\boldsymbol{z})] - D_{\mathrm{KL}}(q\|p)$
- Reparameterisation trick
- Encoder $\rightarrow (\boldsymbol{\mu}, \boldsymbol{\sigma})$; Decoder $\rightarrow \hat{\boldsymbol{x}}$

VAE vs. Standard AE

- Structured, continuous latent space
- Sample: $\boldsymbol{z} \sim \mathcal{N}(0, I)$
- Disentanglement (β-VAE)
- Applications:
  molecular generation,
  anomaly detection,
  data interpolation

Week 14 (Apr 27 – May 1) — Diffusion Models

Denoising Diffusion Probabilistic Models

- Forward: add Gaussian noise gradually
  $q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\,\boldsymbol{x}_{t-1}, \beta_t \boldsymbol{I})$
- Reverse: learn to denoise step-by-step
- Score matching and Langevin dynamics
- Noise schedules: linear, cosine
- Link to stochastic differential equations

Why Diffusion Models?

- State-of-the-art sample quality
- Stable training vs. GANs
- Conditional generation
- Applications:
  molecular structure,
  protein folding,
  lattice field theory

Week 15 (May 4–8) — Diffusion Models (cont.) & GANs

Diffusion: Codes & Implementations

- Implementing DDPM from scratch
- U-Net backbone for the denoiser
- Training loop and ELBO loss
- Sampling procedures
- Classifier-free guidance

Generative Adversarial Networks

Minimax:

$\min_G \max_D \ \mathbb{E}[\log D(\boldsymbol{x})] + \mathbb{E}[\log(1-D(G(\boldsymbol{z})))]$

- Training dynamics and mode collapse
- Wasserstein GAN (WGAN)
- Conditional GANs (cGAN)

Week 16 (May 18–22) — GANs (cont.) & Project Discussions

Advanced GAN Topics

- Architectures: DCGAN, StyleGAN, CycleGAN
- Evaluation: FID, IS, precision/recall
- GANs: fast simulation, inverse problems, lattice QCD
- GANs vs. diffusion: trade-offs

Project Presentations

Applications across the curriculum:

- PINNs for PDEs
- CNNs for detector data
- RBMs as wave-function ansätze
- VAEs for molecular design
- Diffusion / GANs for fast simulation

Weekly Road Map — Part I: Discriminative Methods

Week	Dates	Part	Topics
1	Jan 19–23	Discriminative	Course intro, NN basics, backpropagation, project discussion
2	Jan 26–30	Discriminative	NN, optimisers, regularisation, solving ODEs with NNs
3	Feb 2–6	Discriminative	Physics-Informed Neural Networks (PINNs)
4–5	Feb 9–20	Discriminative	Convolutional NNs: mathematics, architectures
6–7	Feb 23–Mar 6	Discriminative	Recurrent NNs, LSTMs, backprop through time
8–9	Mar 9–20	Discriminative	Autoencoders, PCA, latent space, implementations
10	Mar 23–27	Discriminative	Transformers, self-attention, Part I summary

Weekly Road Map — Part II: Generative Methods

Week	Dates	Part	Topics
11	Apr 6–10	Generative	Generative models overview, Boltzmann machines, MCMC, Gibbs
12	Apr 13–17	Generative	Energy-based models, wave-function ansatz
13	Apr 20–24	Generative	Variational autoencoders (VAEs), ELBO, reparameterisation
14	Apr 27–May 1	Generative	Diffusion models: forward/reverse process, score matching
15	May 4–8	Generative	Diffusion codes (DDPM), GANs: minimax, WGAN, cGAN
16	May 18–22	Generative	Advanced GANs, evaluation metrics, project presentations

What You Will Know at the End

Discriminative Skills

- Derive and implement backpropagation
- Build NNs, CNNs, RNNs, and PINNs
- Solve ODEs and PDEs with neural networks
- Design autoencoders for data reduction
- Understand attention and transformers

Generative Skills

- Train Boltzmann machines and sample from them
- Derive and implement the VAE ELBO
- Implement denoising diffusion models
- Train and stabilise GANs
- Apply generative models to problems in the physical sciences

Overarching Goal

Build both the **mathematical foundations** and the **practical implementation skills** to apply state-of-the-art deep learning methods to open problems in the physical sciences.

Perspectives

Future Directions in Advanced ML

For the Physical Sciences and Beyond

Perspectives — Physics and Chemistry

Physics

- **Many-body quantum systems:** neural-network quantum states (NQS) as variational ansätze for ground and excited states
- **High-energy physics:** jet tagging, event generation, fast detector simulation with GANs and diffusion models
- **Astro- and cosmological data:** field-level inference, gravitational-wave denoising, CMB analysis with score-based generative models
- **Condensed matter:** phase classification

Chemistry

- **Molecular generation:** VAEs, diffusion models, and flow-matching for de-novo drug and catalyst design
- **Property prediction:** graph neural networks (GNNs) on molecular graphs: energies, spectra, reaction rates
- **Quantum chemistry:** ML surrogates for DFT and CCSD(T); PINNs for the electronic Schrödinger equation
- **Reaction pathways:** transformers for retrosynthesis and transition-state prediction, catalysis

Perspectives — Mathematics and Geoscience

Mathematics

- **Operator learning:** DeepONet and Fourier Neural Operators for mesh-free, resolution-independent PDE solvers
- **Inverse problems:** posterior sampling with score-based diffusion; learned regularisers replacing hand-crafted priors
- **Symbolic regression:** discovering equations and conservation laws directly from data
- **Uncertainty quantification:** Bayesian deep learning and conformal prediction for rigorous error bounds

Geoscience

- **Weather and climate:** data-driven emulators (GraphCast, FourCastNet) for fast global forecasts; hybrid physics–ML climate models
- **Seismology:** CNNs and RNNs for earthquake detection, waveform inversion, and induced-seismicity monitoring
- **Remote sensing:** multi-modal transformers for land use, ice-sheet dynamics, and ocean surface reconstruction
- **Subsurface imaging:** reservoir

Perspectives — Bioscience

Structural and Molecular Biology

- **Protein structure prediction:** AlphaFold2 and successors combine transformers with equivariant GNNs; diffusion models (RFDiffusion) for *de-novo* protein design
- **RNA and DNA:** sequence-to-function models, splicing prediction, epigenomics
- **Drug discovery:** generative molecular design, binding affinity prediction, multi-target optimisation

Systems and Medical Biology

- **Single-cell genomics:** VAEs and diffusion for cell-type discovery and trajectory inference
- **Medical imaging:** segmentation and anomaly detection with CNNs and score-based models; synthetic data for rare-disease augmentation
- **Biophysical simulations:** ML force fields (ANI, NequIP) and coarse-grained MD for proteins and membranes

Perspectives — Agentic AI and Reinforcement Learning

Reinforcement Learning for the Physical Sciences

- **Quantum control:** RL agents learn pulse sequences for high-fidelity gate synthesis and robust quantum error correction
- **Experiment design:** active learning with RL for adaptive data acquisition in particle physics and materials science
- **Molecular design:** RL-driven exploration of chemical space guided by property reward functions
- **Plasma and fusion:** real-time magnetic confinement via deep RL policies (DeepMind / ITER)

Agentic AI in Science

- **Autonomous research loops:** LLM-based agents that formulate hypotheses, run simulations, interpret results, and iterate without human intervention
- **Multi-agent systems:** specialised agents for literature search, code generation, data analysis, and experimental planning collaborate in a shared scientific workflow
- **Human-in-the-loop science:** agentic pipelines as co-pilots for accelerating discovery in drug

Perspectives — Graph Networks and Foundation Models

Graph Neural Networks

- Natural representation of molecules, crystals, interaction networks, and complex systems
- **Equivariant GNNs** (SE(3)-Transformers, NequIP, MACE) enforce physical symmetries exactly
- Applications: molecular force fields, particle-physics event reconstruction, social and biological networks
- **Open challenge:** scaling equivariant architectures to very large systems efficiently

Foundation Models and Transfer Learning

- Large pre-trained models (ESM-2, ChemBERTa, ClimaX, AstroCLIP) transfer to downstream science tasks with little labelled data
- **Scientific LLMs:** models fine-tuned on papers, code, and experimental data for literature synthesis and hypothesis generation
- **Multi-modal learning:** joint text, structure, spectrum, and image representations for universal scientific models